

ANUM AAMIR

Full-Stack Developer | AI & QA Enthusiast

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PROFILE

Computer Science undergraduate with full-stack development experience (Node.js, Express, React, Python) and hands-on exposure to QA and testing. Comfortable across backend, frontend, and databases, with a growing interest in AI/ML and a strong record of academic achievement.

WORK EXPERIENCE

Flutter Development Intern — NextGen

August 2026 – Present

- Built foundational Flutter UI screens and reusable custom widgets using Scaffold, Row/Column, ListView, and GridView.
- Implemented app navigation (Navigator/Routes, Bottom Navigation Bar) and a CRUD To-Do app, migrating state management from setState() to Provider.
- Integrated REST APIs in a Weather App using JSON parsing, loading indicators, and error handling.

Offensive Security Intern — Itsolera, Lahore

June 2026 – August 2026

- Collaborated with a 6-person team to build a modular Python CLI reconnaissance framework automating passive/active recon (WHOIS, DNS, subdomain enumeration, port scanning) for penetration testing engagements.
- Built a flag-driven web security testing framework in Python automating detection of common vulnerabilities (XSS, SQL injection, authentication weaknesses).
- Replicated a known CVE in an isolated lab and documented root-cause analysis and remediation steps, contributing to a 648-line pytest test suite and a Dockerized deployment.

SQA Intern — Dafi Labs, Lahore

July 2026 – August 2026

- Performed manual and automation testing across web and application platforms, including test case design, execution, and defect reporting (STLC/SDLC).
- Built and debugged automated UI test suites in Playwright and Cypress (Page Object Model) across Chromium and Firefox, plus equivalent suites in Selenium and Robot Framework, with automated HTML/Mochawesome reporting.
- Designed a Postman API test collection with chained requests via environment variables, covering GET/POST/PUT/DEL.

Software Development Intern — Pentaloop (Piron Labs), Lahore

July 2025

- Built and maintained a Node.js/Express backend with custom routing and RESTful API endpoints; developed a Python automation pipeline for keyword scraping using BeautifulSoup, Requests, and Selenium.
- Wrote modular, reusable JavaScript and Python code following team coding standards; validated and debugged API endpoints using Postman and Thunder Client alongside the development team.

EDUCATION

FAST NUCES Lahore

BS Computer Science
2023 – Present

Lahore Grammar School

O Levels 2018 – 2021
A Levels 2021 – 2023

SKILLS

Languages: C, C++, Python, JavaScript, Assembly Language, SQL

Web and Backend: HTML, CSS, Node.js, Express.js, React, REST APIs

QA and Testing: Manual Testing, Automation Testing, Web and Application Testing, Functional/Regression/UI Testing, Page Object Model, Cross-Browser Testing, Keyword-Driven Testing

Automation and Scraping: BeautifulSoup, Requests, Selenium

AI and Machine Learning: Deep Learning, RNN, CNN, TensorFlow/Keras

Testing and QA Tools: SQA Fundamentals, Unit/Integration/System Testing, API Testing, Postman, Thunder Client, Selenium, Playwright, Cypress, Robot Framework

Tools: VS Code, Git, Postman, Jira, Figma

ACHIEVEMENTS

- Recipient of the Dean's Certificate for Fall 2023 and Spring 2026
- A Levels: 100% Merit Scholarship recipient
- O Levels: Achieved 9 A*/A grades
- O Levels: 2nd position internationally, CEATS Mathematics Competition (2020)
- O Levels: Awarded the High Achiever Gold Medal (2021)

PROJECTS

InSync – AI-Powered Collaborative Learning Platform (Final Year Project)

2026 – Present

React/Next.js, Node.js/Express, FastAPI, PostgreSQL, Pinecone, WebSocket/Socket.IO

- Building a multi-agent AI system (Tutor, Quiz, Progress, Resource, Teacher Assistant) with real-time teacher dashboards and a live collaborative BST whiteboard/coding workspace.
- Early testing shows a 20% improvement in student understanding of BST concepts.

Umeed – AI-Powered Mental Health First-Aid App (Rescue Hacks 2026 Hackathon)

2026

Next.js, TypeScript, Anthropic Claude API, Node.js/Express, pdf-lib

- Built a Claude-powered mental health chat app with an independent, server-side crisis-detection layer tested against 20+ phrasings, plus prompt guardrails validated by an automated scoring harness (8 scenarios, avg 4.0/5).
- Added a PDF-exportable, AI-co-authored safety-plan builder and preference-based (non-demographic) crisis resource matching.

Terrarium – Climate Intervention Digital Twin (Case Study)

2026

Python, LightGBM, FFT Dispersion Modelling, WorldPop, Landsat, OpenStreetMap, LLM Integration

- Built a neighbourhood-scale climate digital twin fusing 7 real-world data sources onto a unified 100m grid; trained a LightGBM emulator and an FFT-based PM2.5 dispersion model validated against 53 real air-quality monitors.
- Delivered an interactive 3D sandbox with LLM-generated result briefs and an equity layer breaking results down by population density decile.

Vanguard Software Solutions – Software House Website

2026

React, Vite, Tailwind CSS, Framer Motion

- Built and animated a marketing site (Home, About, Services, Contact) with Framer Motion transitions, optimised for lead generation across desktop and mobile.

Kebabistan – Restaurant Ordering and Admin Platform

2026

JavaScript, Node.js, Express.js, SQLite

- Full-stack restaurant platform with live ordering, table reservations, and a secure admin dashboard for real-time revenue, stock, and booking management.
- Express.js backend with an auto-seeding SQLite database and GSAP-powered frontend animations.

Reelify – Automated Reel Generator

2026

React, Express.js

- Automated video-reel generator: Gemini/HuggingFace for image generation, FFmpeg to stitch images into captioned, scored videos.

Research Paper Implementation: Motion & Object Detection

2026

Python, CNN, Deep Learning

- Team-based AI semester project: reproduced a published CNN/RNN research paper for object and human-activity detection, benchmarking results against the original outcomes.

Society Management Website

2025

Node.js, React, SQL

- Full-stack platform for managing university societies, with a relational database schema and interfaces for membership, events, and budget tracking.
- Built secure JWT authentication and communication tools, translating real admin and member requirements into a practical platform.